

**JAYPEE UNIVERSITY OF ENGINEERING & TECHNOLOGY, GUNA**  
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

Course: Computer Organization & Architecture Lab

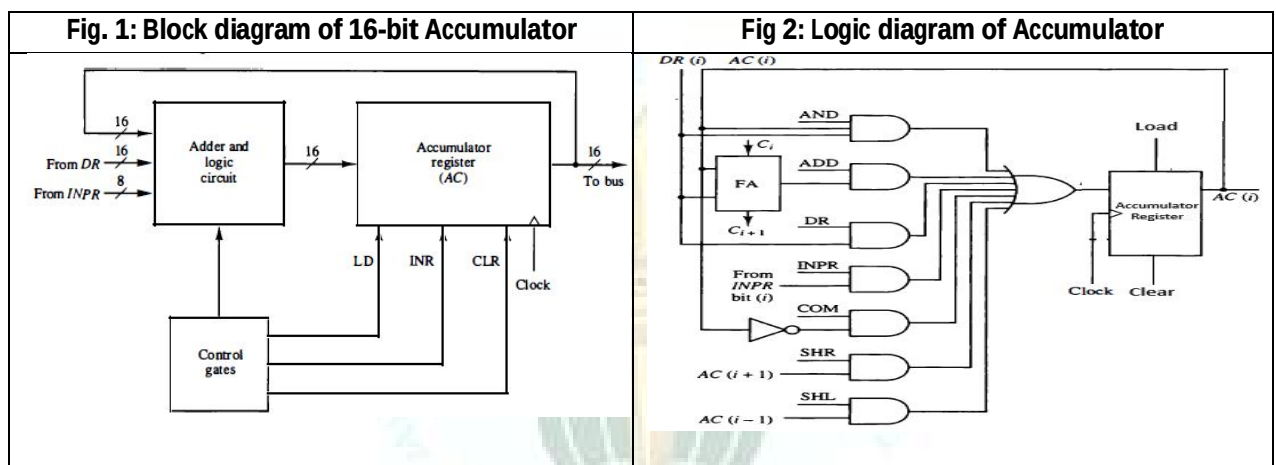
Course Code: CS208

B. Tech. (CSE IV Sem.)

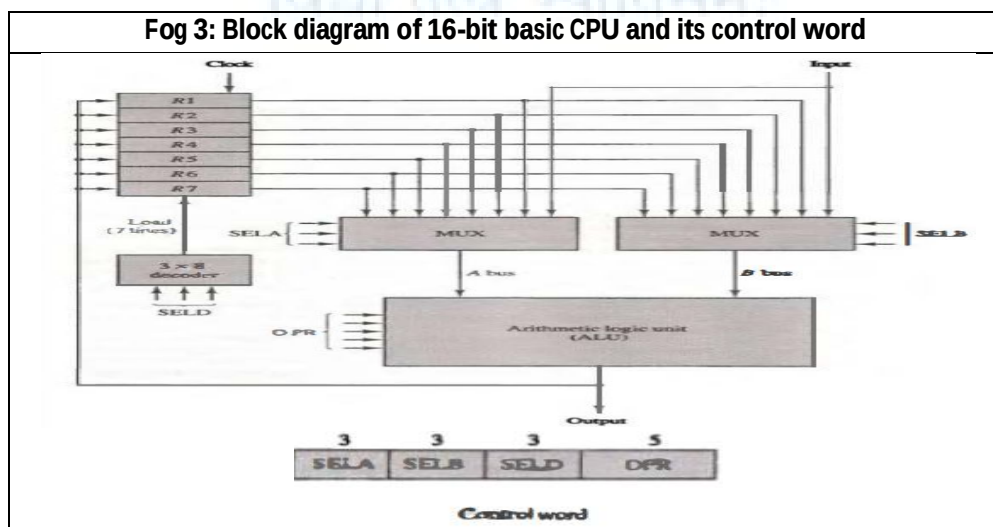
**Experiment # 9**

**Aim: Design of accumulator and basic CPU.**

**Accumulator:** This is a register used to store the intermediate results of all arithmetic, logical shift operations. It plays a crucial role in performing calculations and processing data in various applications.



**Central Processing Unit:** This is the brain of the computer that performs the bulk of data-processing operations and is referred to as the CPU. The CPU is made up of three major parts, the register set stores intermediate data used during the execution of the instructions, arithmetic logic unit (ALU) performs the required micro operations for executing the instructions and the control unit that supervises the transfer of information among the registers and instructs the ALU as to which operation to perform.



**Exercise#1:** Design and verify 8-bit logic diagram of accumulator shown Fig. 1 in that performs following operations. Use **in built blocks** of register as accumulator, 8:1 multiplexer as control (gates) unit, and two 8-bit input pins one as data register (DR) and other as input register (INPR).

|                               |                    |
|-------------------------------|--------------------|
| $AC \leftarrow AC \wedge DR$  | AND with DR        |
| $AC \leftarrow AC + DR$       | Add with DR        |
| $AC \leftarrow DR$            | Transfer from DR   |
| $AC(0-7) \leftarrow INPR$     | Transfer from INPR |
| $AC \leftarrow \overline{AC}$ | Complement         |
| $AC \leftarrow shr AC,$       | Shift right        |
| $AC \leftarrow shl AC,$       | Shift left         |
| $AC \leftarrow 0$             | Clear              |
| $AC \leftarrow AC + 1$        | Increment          |

**Exercise#2:** Design and verify 4-bit CPU shown Fig. 3 that performs operations given in Table-2. Use **in built blocks** of registers, multiplexers and decoder and **sub circuit** of ALU (designed in experiment no. 6).

| Encoding of Register Selection Fields |       |       |      | Encoding of ALU Operations |                                                |
|---------------------------------------|-------|-------|------|----------------------------|------------------------------------------------|
| Binary Code                           | SELA  | SELB  | SELD | OPR Select                 | Arithmetic Operations                          |
| 000                                   | Input | Input | None | 000                        | Transfer : $F = A$                             |
| 001                                   | R1    | R1    | R1   | 001                        | Subtract: $F = A - B \ (A + \overline{B} + 1)$ |
| 010                                   | R2    | R2    | R2   | 010                        | Addition: $F = A + B$                          |
| 011                                   | R3    | R3    | R3   | 011                        | Decrement: $F = A - 1$                         |
| 100                                   | R4    | R4    | R4   | <b>Logical Operations</b>  |                                                |
| 101                                   | R5    | R5    | R5   | 100                        | XOR: $F = A \oplus B$                          |
| 110                                   | R6    | R6    | R6   | 101                        | AND: $F = A \wedge B$                          |
| 111                                   | R7    | R7    | R7   | <b>Shift Operations</b>    |                                                |
|                                       |       |       |      | 110                        | Circular left shift of A: $F = cil A$          |
|                                       |       |       |      | 111                        | Arithmetic shift of B: $F = Ashr B$            |